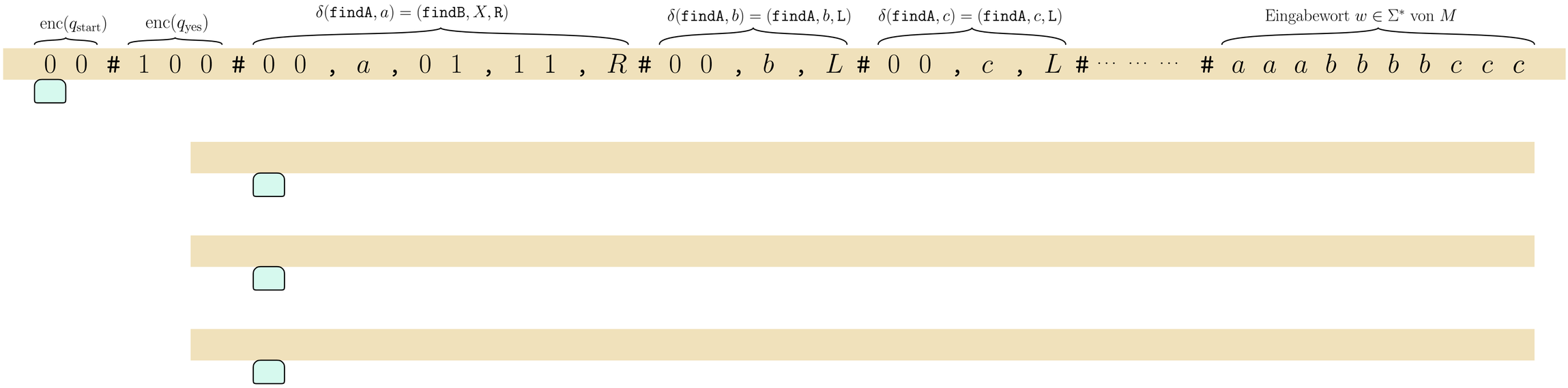
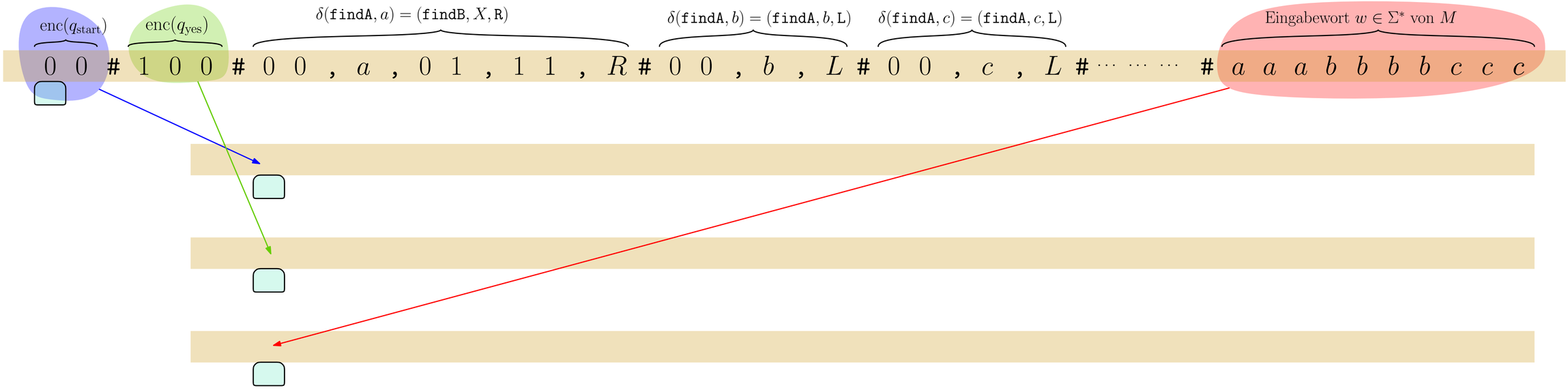
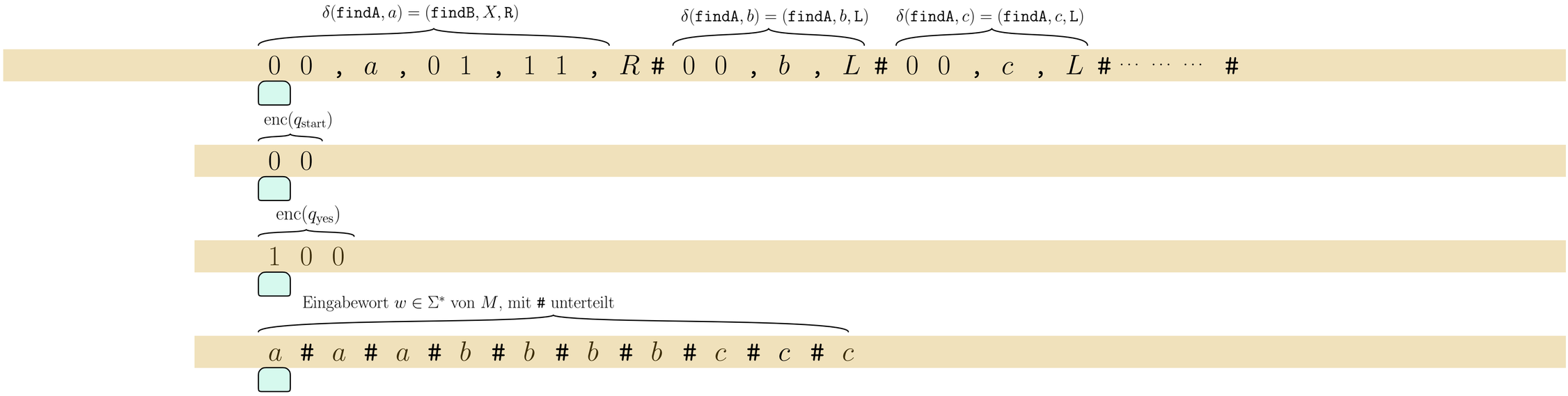
0 0 # 1 0 0 # 0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # $\cdots\cdots\cdots$ # a a a b b b b c c c

$$\overbrace{0\ 0}^{\mathrm{enc}(q_{\mathrm{start}})}\ \#\ \overbrace{1\ 0\ 0}^{\mathrm{enc}(q_{\mathrm{yes}})}\ \#\ \overbrace{0\ 0\ ,\ a\ ,\ 0\ 1\ ,\ 1\ 1\ ,\ R}^{\delta(\texttt{findA},a)=(\texttt{findB},X,\mathtt{R})}\ \#\ \overbrace{0\ 0\ ,\ b\ ,\ L}^{\delta(\texttt{findA},b)=(\texttt{findA},b,\mathtt{L})}\ \#\ \overbrace{0\ 0\ ,\ c\ ,\ L}^{\delta(\texttt{findA},c)=(\texttt{findA},c,\mathtt{L})}\ \#\ \cdots\ \cdots\ \cdots\ \#\ \overbrace{a\ a\ a\ b\ b\ b\ b\ c\ c\ c}^{\text{Eingabewort } w\in\Sigma^* \text{ von } M}$$


enc(q_{start})
enc(q_{yes})
$\delta(\texttt{findA}, a) = (\texttt{findB}, X, \texttt{R})$
$\delta(\texttt{findA}, b) = (\texttt{findA}, b, \texttt{L})$
$\delta(\texttt{findA}, c) = (\texttt{findA}, c, \texttt{L})$
Eingabewort $w \in \Sigma^*$ von M
0 0 # 1 0 0 # 0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· # a a a b b b b c c c


enc(q_{start})
enc(q_{yes})
$\delta(\texttt{findA}, a) = (\texttt{findB}, X, \texttt{R})$
$\delta(\texttt{findA}, b) = (\texttt{findA}, b, \texttt{L})$
$\delta(\texttt{findA}, c) = (\texttt{findA}, c, \texttt{L})$
Eingabewort $w \in \Sigma^*$ von M
0 0 # 1 0 0 # 0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· # a a a b b b b c c c


$\delta(\texttt{findA}, a) = (\texttt{findB}, X, \texttt{R})$
$\delta(\texttt{findA}, b) = (\texttt{findA}, b, \texttt{L})$
$\delta(\texttt{findA}, c) = (\texttt{findA}, c, \texttt{L})$
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
$\text{enc}(q_{\text{start}})$
0 0
$\text{enc}(q_{\text{yes}})$
1 0 0
Eingabewort $w \in \Sigma^*$ von M, mit # unterteilt
a # a # a # b # b # b # b # c # c # c

Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 0
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c

Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 0
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c

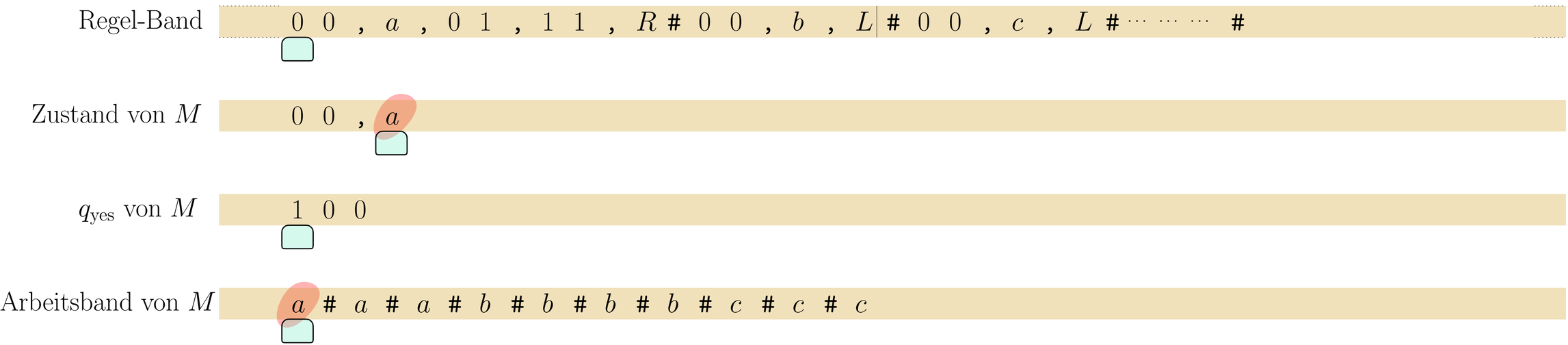
Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 0
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c

Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 0
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c

Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 0
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c

Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 0 ,
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c

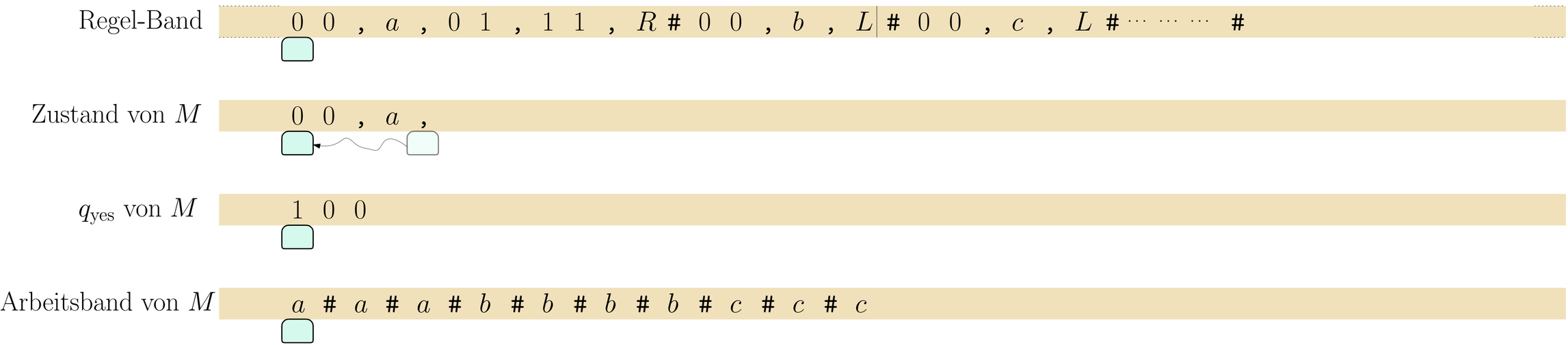
Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 0 ,
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c


Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
Zustand von M
0 0 , a
q_{yes} von M
1 0 0
Arbeitsband von M
a # a # a # b # b # b # b # c # c # c

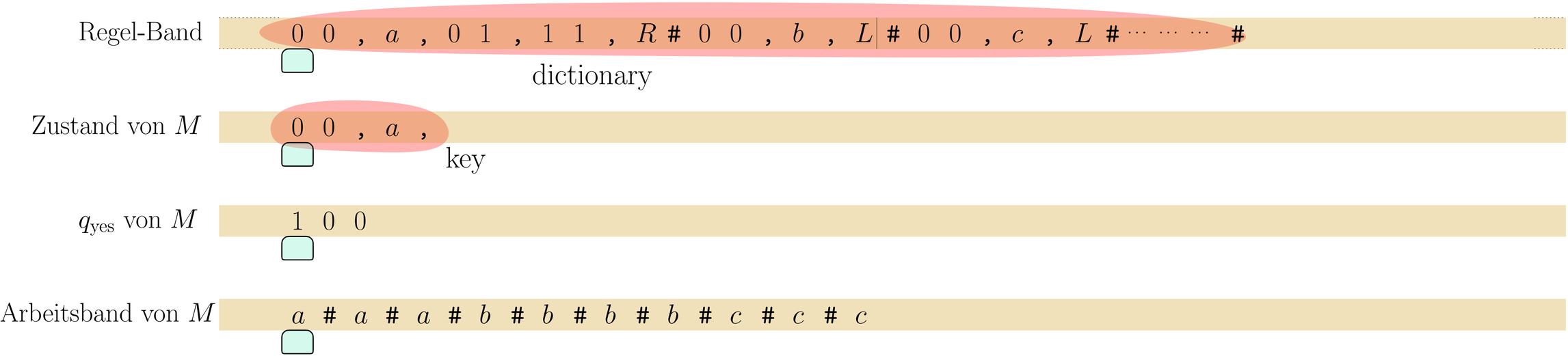
Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 0 , a
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c

Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 0 , a
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c

Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 0 , a ,
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c


Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
Zustand von M
0 0 , a ,
q_yes von M
1 0 0
Arbeitsband von M
a # a # a # b # b # b # b # c # c # c

Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 0 , a ,
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c


Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
dictionary
Zustand von M
0 0 , a ,
key
q_yes von M
1 0 0
Arbeitsband von M
a # a # a # b # b # b # b # c # c # c

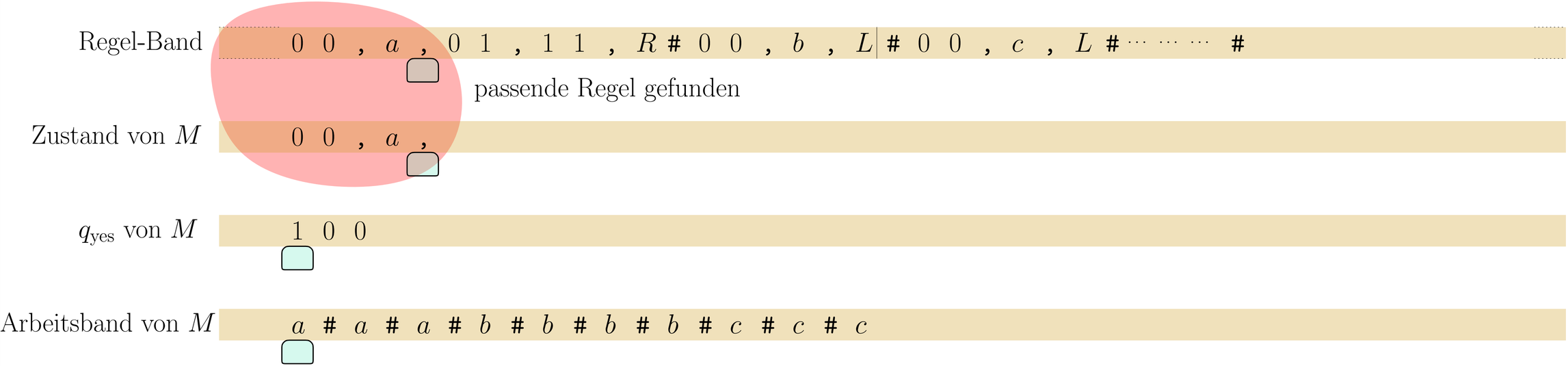
Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 0 , a ,
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c

Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 0 , a ,
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c

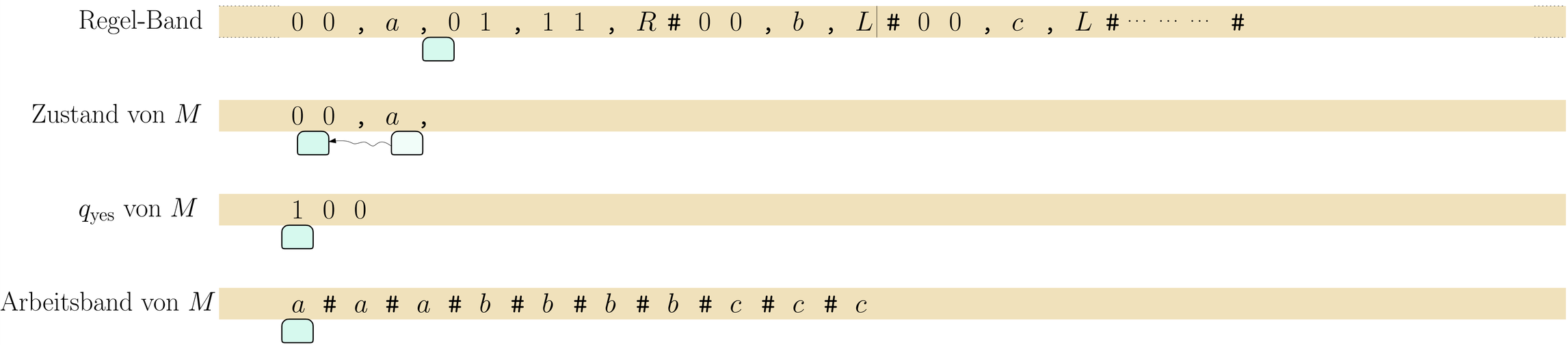
Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 0 , a ,
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c

Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 0 , a ,
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c

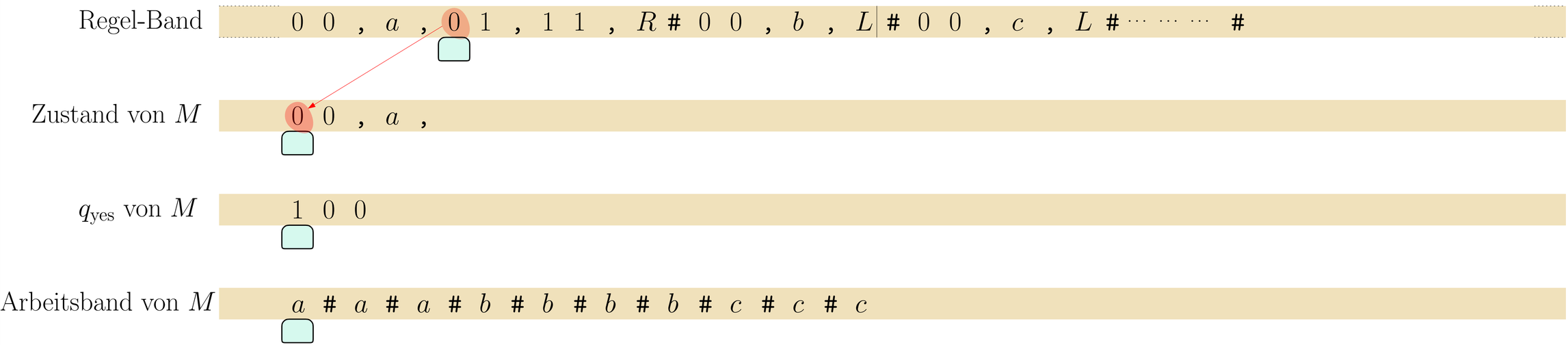
Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 0 , a ,
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c

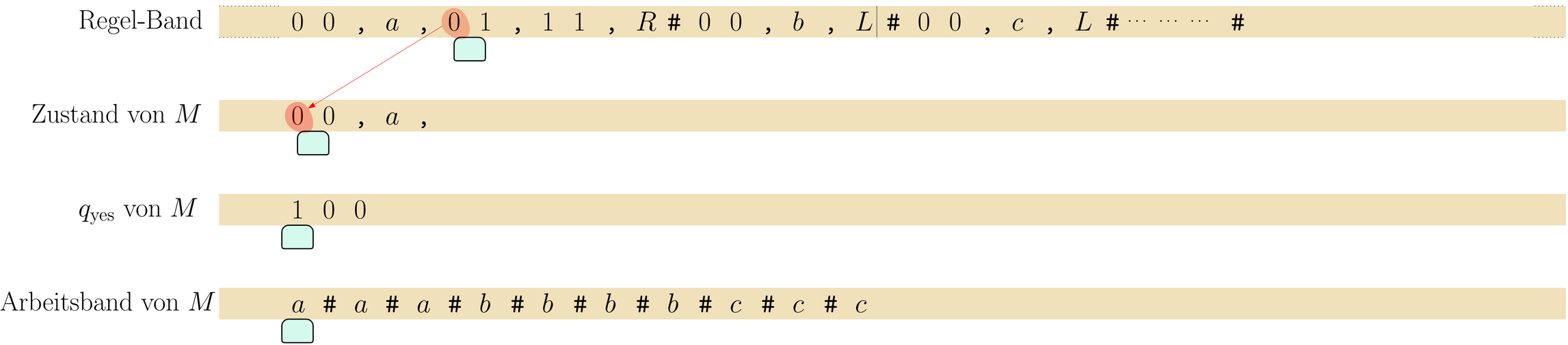

Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
passende Regel gefunden
Zustand von M
0 0 , a ,
q_yes von M
1 0 0
Arbeitsband von M
a # a # a # b # b # b # b # c # c # c

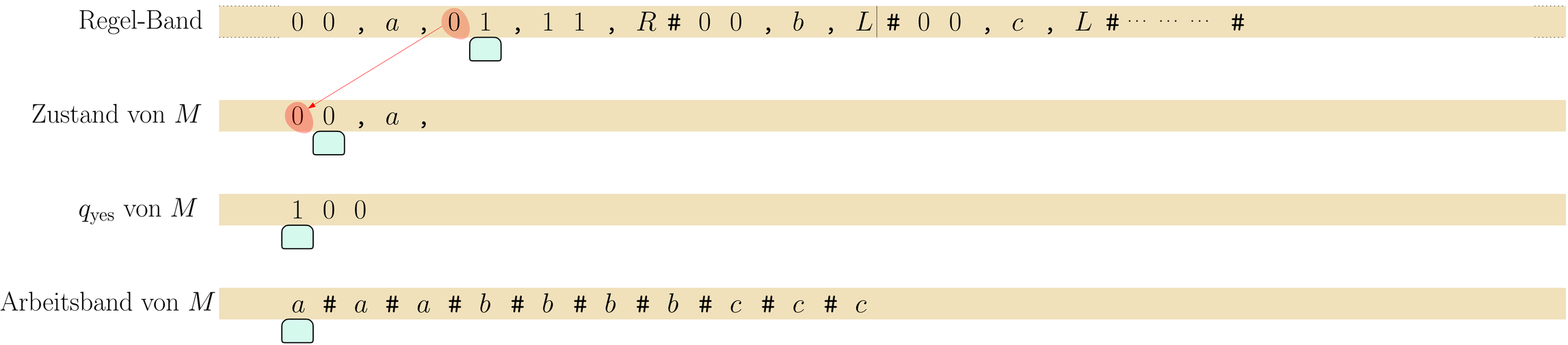
Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 0 , a ,
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c

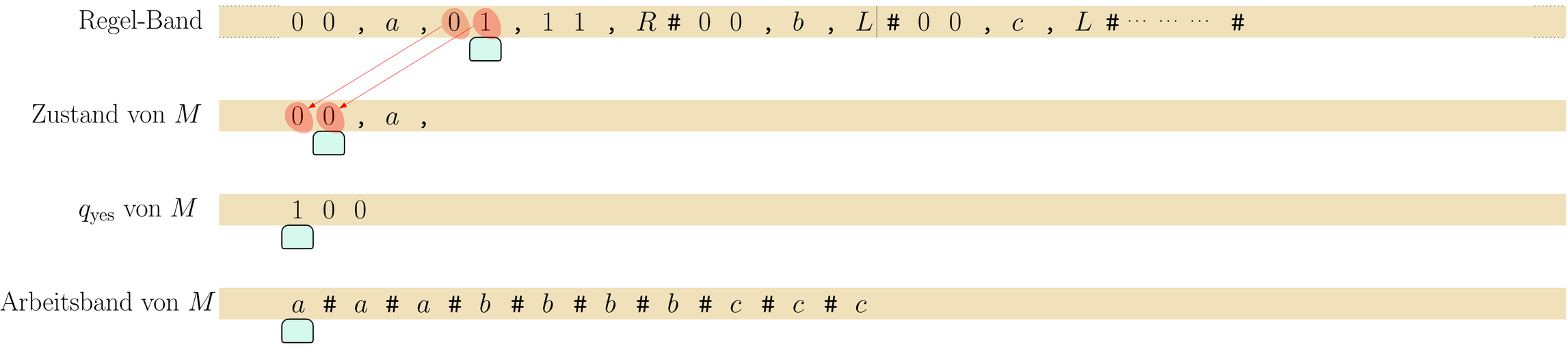

Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
Zustand von M
0 0 , a ,
q_yes von M
1 0 0
Arbeitsband von M
a # a # a # b # b # b # b # c # c # c

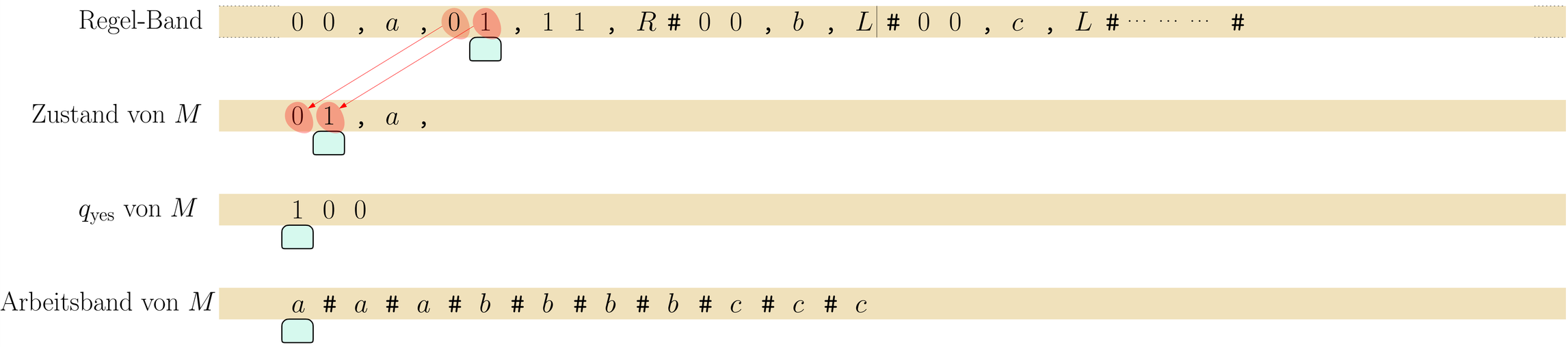
Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 0 , a ,
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c


Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
Zustand von M
0 0 , a ,
q_{yes} von M
1 0 0
Arbeitsband von M
a # a # a # b # b # b # b # c # c # c


Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
Zustand von M
0 0 , a ,
q_yes von M
1 0 0
Arbeitsband von M
a # a # a # b # b # b # b # c # c # c


Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
Zustand von M
0 0 , a ,
q_yes von M
1 0 0
Arbeitsband von M
a # a # a # b # b # b # b # c # c # c


Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
Zustand von M
0 0 , a ,
q_yes von M
1 0 0
Arbeitsband von M
a # a # a # b # b # b # b # c # c # c


Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
Zustand von M
0 1 , a ,
q_yes von M
1 0 0
Arbeitsband von M
a # a # a # b # b # b # b # c # c # c

Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 1 , a ,
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c

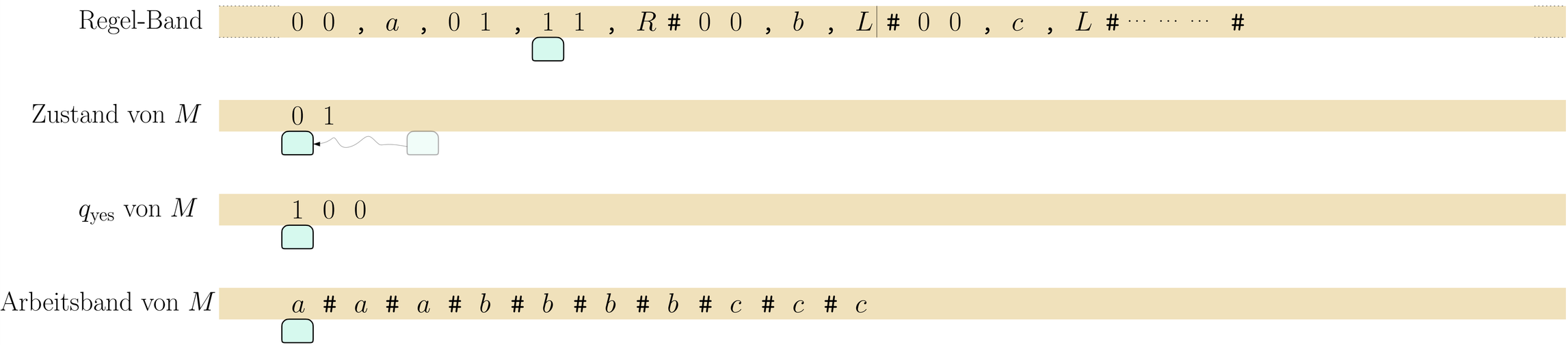
Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 1 , a ,
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c

Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 1 a ,
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c

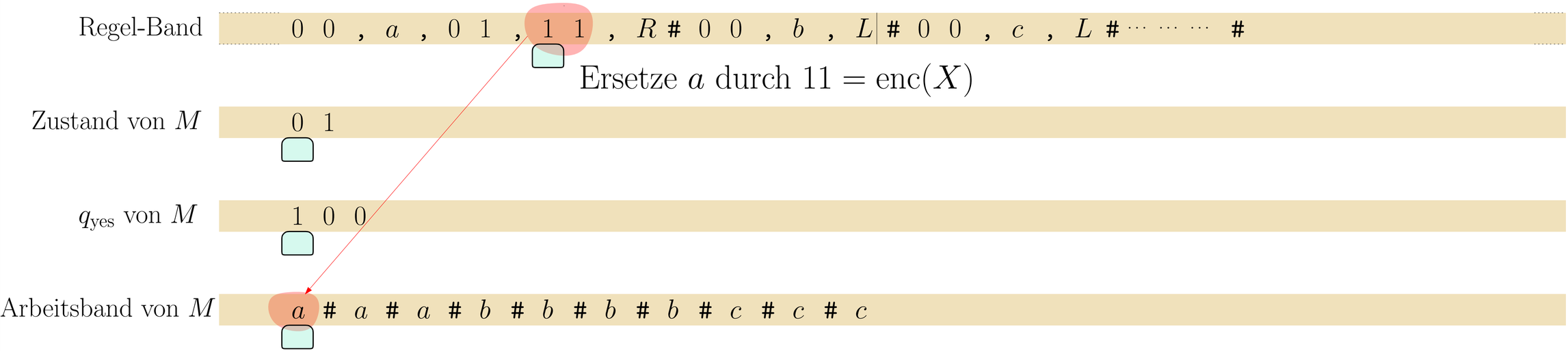
Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 1 a ,
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c

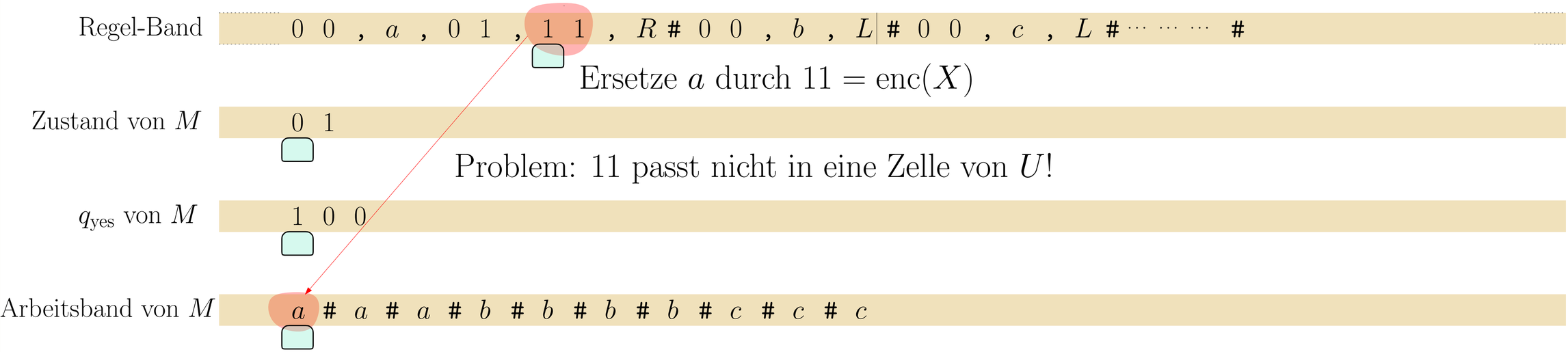
Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 1 ,
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c

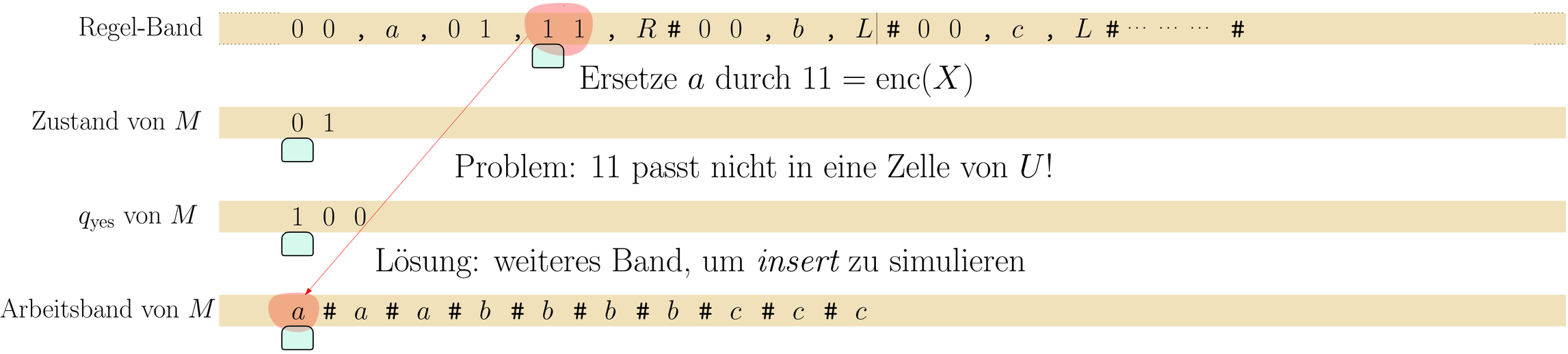
Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 1 ,
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c


Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
Zustand von M
0 1
q_{yes} von M
1 0 0
Arbeitsband von M
a # a # a # b # b # b # b # c # c # c

Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 1
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c


Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
Ersetze a durch 11 = enc(X)
Zustand von M
0 1
q_yes von M
1 0 0
Arbeitsband von M
a # a # a # b # b # b # b # c # c # c


Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
Ersetze a durch 11 = enc(X)
Zustand von M
0 1
Problem: 11 passt nicht in eine Zelle von U!
q_yes von M
1 0 0
Arbeitsband von M
a # a # a # b # b # b # b # c # c # c


Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
Ersetze a durch 11 = enc(X)
Zustand von M
0 1
Problem: 11 passt nicht in eine Zelle von U!
q_yes von M
1 0 0
Lösung: weiteres Band, um insert zu simulieren
Arbeitsband von M
a # a # a # b # b # b # b # c # c # c

Regel-Band: 0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ··· ··· ··· #

Ersetze a durch $11 = \mathrm{enc}(X)$

Zustand von M: 0 1

Problem: 11 passt nicht in eine Zelle von U!

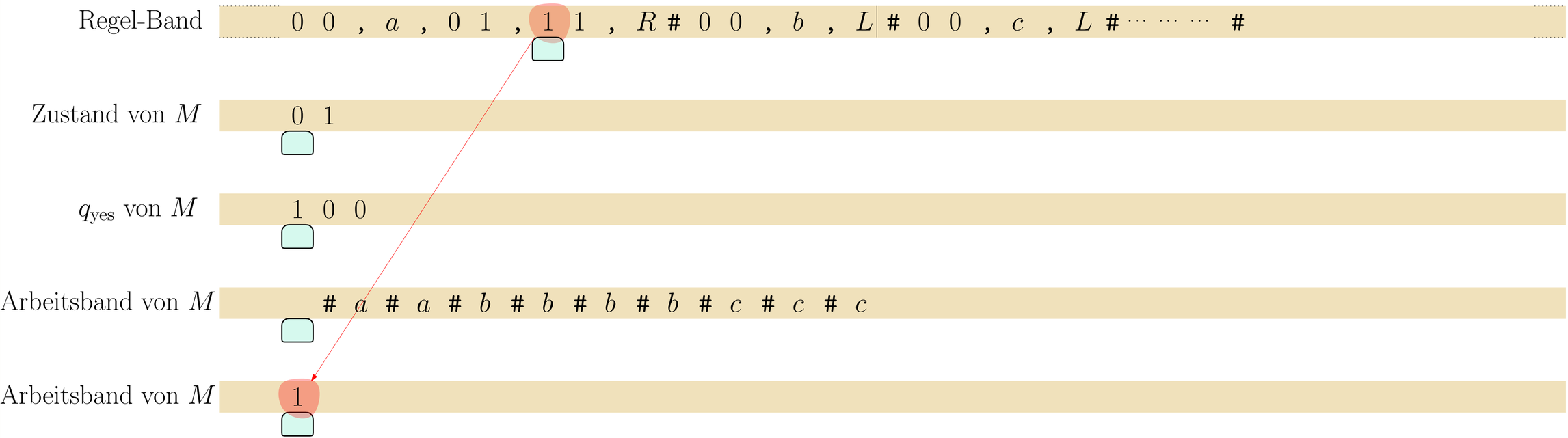
q_{yes} von M: 1 0 0

Lösung: weiteres Band, um *insert* zu simulieren

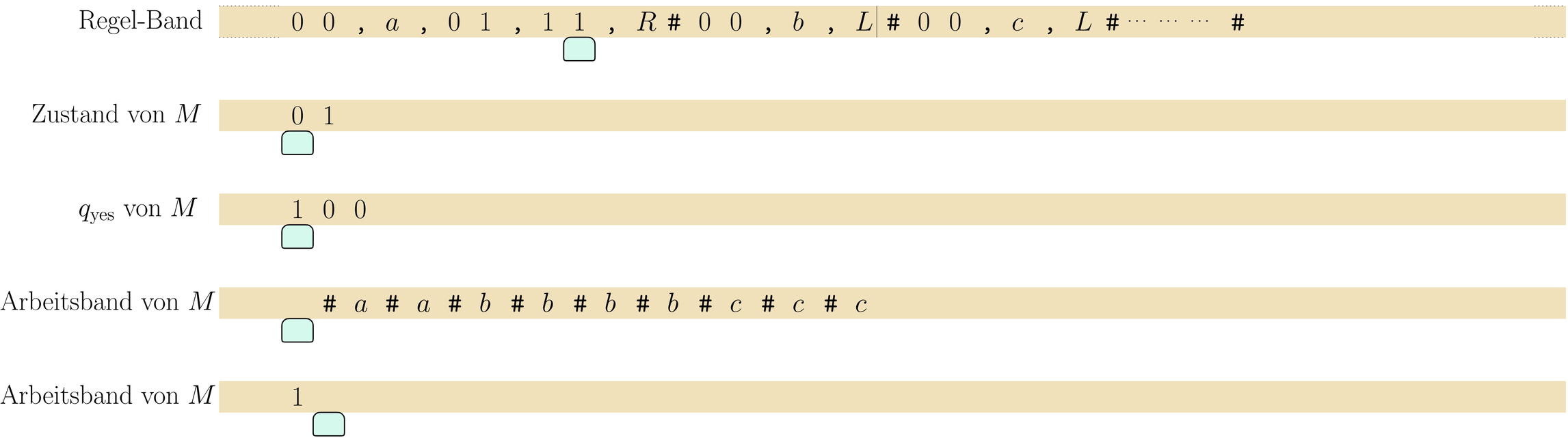
Arbeitsband von M: a # a # a # b # b # b # b # c # c # c

Arbeitsband von M:

Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 1
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # a # b # b # b # b # c # c # c
Arbeitsband von M	

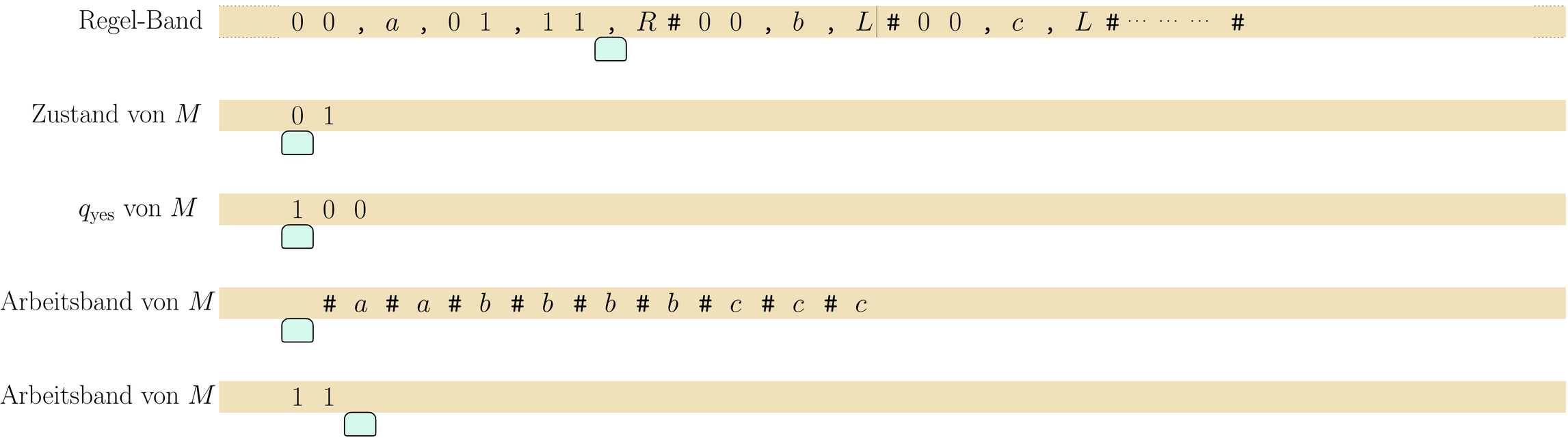

Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
Zustand von M
0 1
q_yes von M
1 0 0
Arbeitsband von M
a # a # b # b # b # b # c # c # c
Arbeitsband von M
1

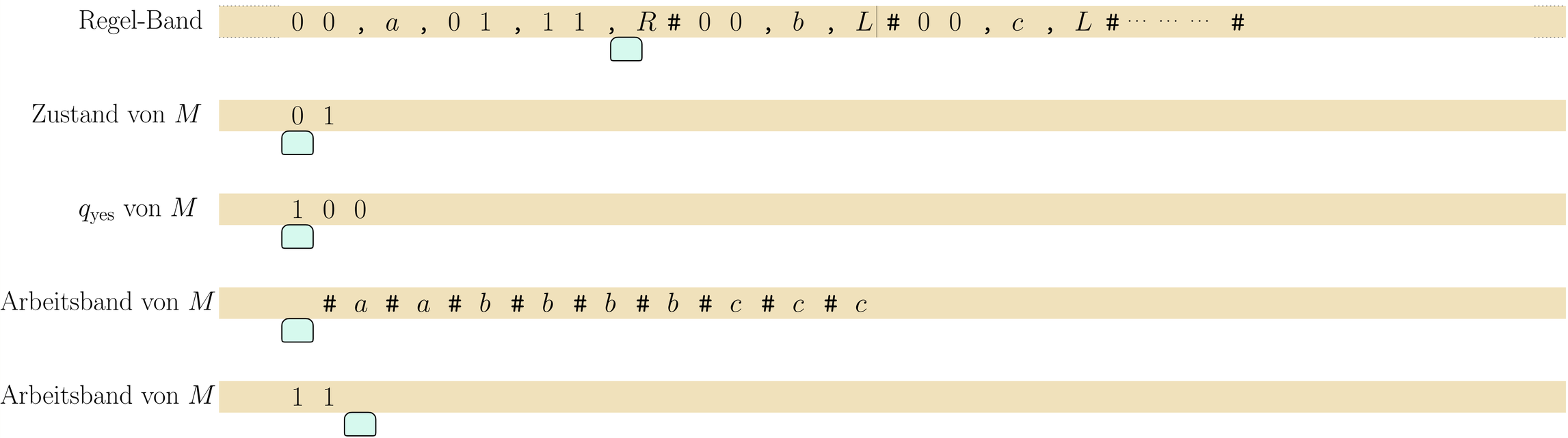
Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 1
q_{yes} von M	1 0 0
Arbeitsband von M	# a # a # b # b # b # b # c # c # c
Arbeitsband von M	1


Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
Zustand von M
0 1
q_yes von M
1 0 0
Arbeitsband von M
a # a # b # b # b # b # c # c # c
Arbeitsband von M
1

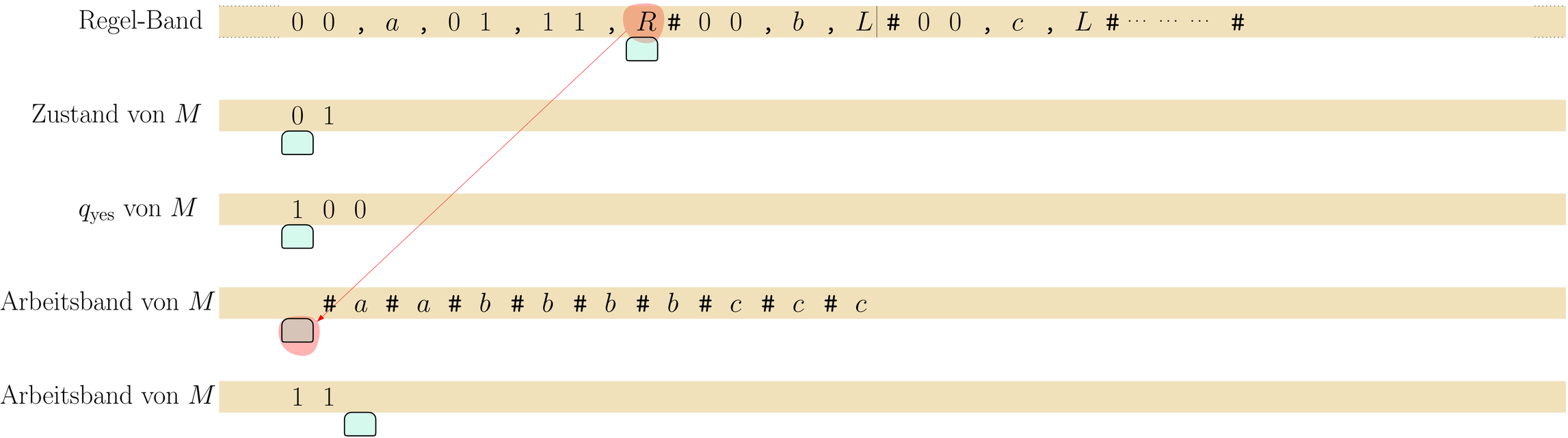
Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 1
q_{yes} von M	1 0 0
Arbeitsband von M	# a # a # b # b # b # b # c # c # c
Arbeitsband von M	1 1

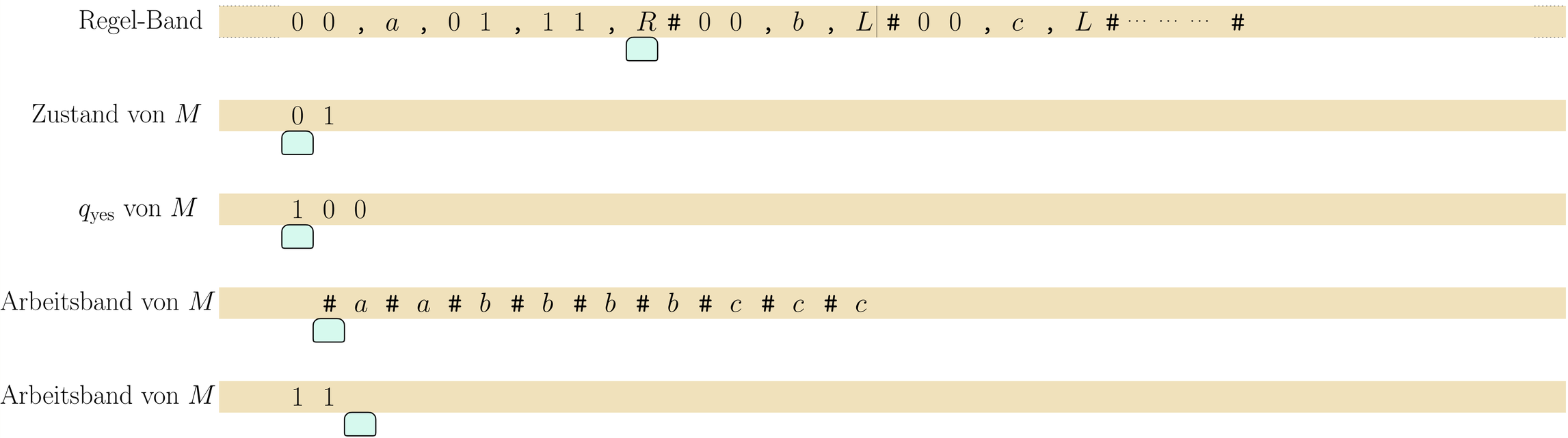
Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ··· ··· ··· #
Zustand von M	0 1
q_{yes} von M	1 0 0
Arbeitsband von M	# a # a # b # b # b # b # c # c # c
Arbeitsband von M	1 1


Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
Zustand von M
0 1
q_yes von M
1 0 0
Arbeitsband von M
a # a # b # b # b # b # c # c # c
Arbeitsband von M
1 1

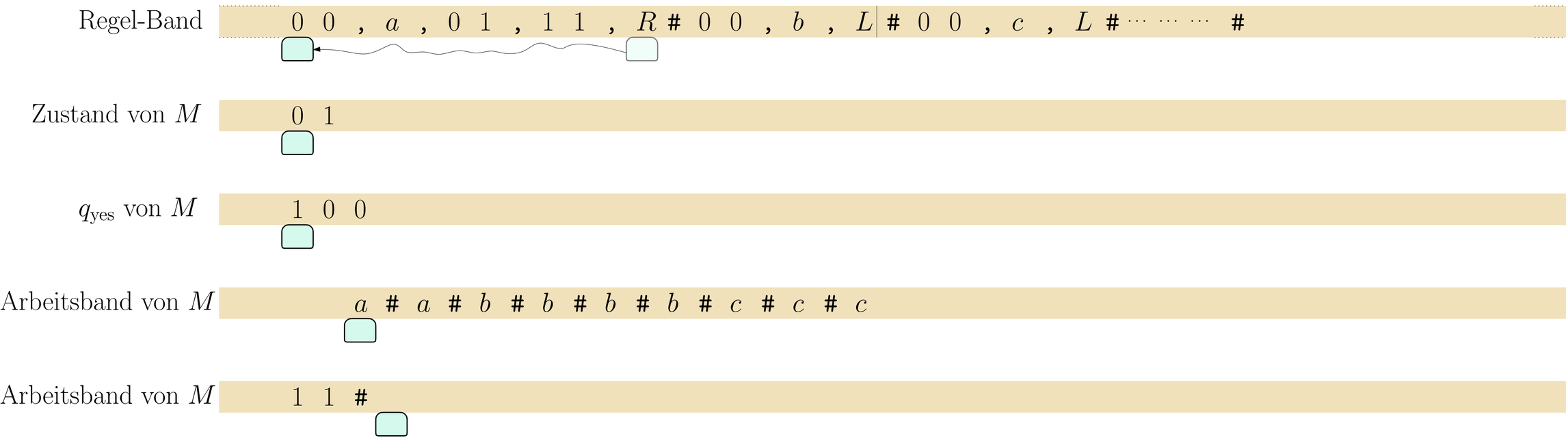

Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
Zustand von M
0 1
q_yes von M
1 0 0
Arbeitsband von M
a # a # b # b # b # b # c # c # c
Arbeitsband von M
1 1

Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ··· ··· ··· #
Zustand von M	0 1
q_{yes} von M	1 0 0
Arbeitsband von M	# a # a # b # b # b # b # c # c # c
Arbeitsband von M	1 1


Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
Zustand von M
0 1
q_yes von M
1 0 0
Arbeitsband von M
a # a # b # b # b # b # c # c # c
Arbeitsband von M
1 1


Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
Zustand von M
0 1
q_yes von M
1 0 0
Arbeitsband von M
a # a # b # b # b # b # c # c # c
Arbeitsband von M
1 1

Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ··· ··· ··· #
Zustand von M	0 1
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # b # b # b # b # c # c # c
Arbeitsband von M	1 1 #


Regel-Band
0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L # 0 0 , c , L # ··· ··· ··· #
Zustand von M
0 1
q_yes von M
1 0 0
Arbeitsband von M
a # a # b # b # b # b # c # c # c
Arbeitsband von M
1 1 #

Regel-Band	0 0 , a , 0 1 , 1 1 , R # 0 0 , b , L | # 0 0 , c , L # ⋯ ⋯ ⋯ #
Zustand von M	0 1
q_{yes} von M	1 0 0
Arbeitsband von M	a # a # b # b # b # b # c # c # c
Arbeitsband von M	1 1 #